

Renal Reference — IV Fluid Types

0.9% Normal Saline — default volume replacement. Avoid when patient already has hyperchloremic metabolic acidosis (it worsens it).

Ringer's Lactate / Hartmann's / Plasma-Lyte — buffered crystalloids, slightly alkaline. Preferred when you want to avoid chloride load. Used in most surgical/ICU volume replacement. Avoid in hyperkalemia (RL contains K).

D5W + NaHCO₃ (bicarbonate solution) — used in rhabdomyolysis to alkalinize urine and prevent myoglobin-Tamm-Horsfall cast formation. Also used when metabolic acidosis needs correction.

Packed RBCs — severe acute blood loss. Not a crystalloid — used when hemorrhage is the primary problem.

Colloids (albumin) — HRS: albumin infusion as bridge therapy. Not used routinely in AKI (hydroxyethyl starch is now off the market — increased AKI risk).

The rule across all of them: **match the fluid to what was lost and to the acid-base status of the patient.**

1. 0.9% Normal Saline

WHAT YOU SEE

0.9% NS — isotonic crystalloid, default volume replacement

WHAT IT ENCODES

0.9% NaCl contains Na⁺ 154 and Cl⁻ 154 mEq/L with osmolarity ~308 — isotonic, distributes through the extracellular space (~1/3 stays intravascular, so ~3 mL given per 1 mL of intravascular expansion). First-line for hypovolemic resuscitation, hypercalcemia (saline diuresis), and hyponatremia volume repletion. The supraphysiologic chloride (plasma Cl⁻ is ~100) is its weakness.

BOARD TRAP

Normal saline is the universally 'safe' default fluid. WRONG — large volumes cause a hyperchloremic non-anion-gap metabolic acidosis and chloride-mediated renal afferent vasoconstriction; balanced crystalloids (LR/Plasma-Lyte) are preferred for most resuscitation (SMART/SALT-ED trials showed fewer major adverse kidney events).

2. Avoid NS in hyperchloremic acidosis

WHAT YOU SEE

Avoid when patient already has hyperchloremic metabolic acidosis (it worsens it)

WHAT IT ENCODES

Each liter of NS adds a 54 mEq/L chloride excess over plasma; by the Stewart strong-ion approach, rising plasma Cl⁻ lowers the strong-ion difference and drops bicarbonate → worsening NAGMA. Avoid in DKA recovery, diarrheal/RTA acidosis, and any established hyperchloremic acidosis; reach for LR/Plasma-Lyte instead.

BOARD TRAP

Give more NS to 'flush out' an existing hyperchloremic acidosis. WRONG — adding chloride deepens the NAGMA; switch to a balanced/buffered crystalloid whose lactate or acetate is metabolized to bicarbonate.

3. Ringer's Lactate / Plasma-Lyte

WHAT YOU SEE

RL / Hartmann's / Plasma-Lyte — buffered crystalloids, slightly alkaline

WHAT IT ENCODES

Lactated Ringer's: Na⁺ 130, Cl⁻ 109, K⁺ 4, Ca²⁺ 3, lactate 28 (hepatically converted to bicarbonate). Plasma-Lyte: Na⁺ 140, Cl⁻ 98, K⁺ 5, Mg²⁺, with acetate/gluconate buffers and no calcium. Both have near-physiologic chloride, so they avoid hyperchloremic acidosis and are preferred in surgical/ICU resuscitation, sepsis, and pancreatitis.

BOARD TRAP

LR is contraindicated in lactic acidosis because it 'adds lactate.' WRONG — the lactate is metabolized to bicarbonate and does not meaningfully raise measured lactate; the real cautions are hyperkalemia (LR has K⁺) and co-infusion with ceftriaxone or blood through the same line (calcium → precipitation/clot risk).

4. LR contains potassium

WHAT YOU SEE

Used in most surgical/ICU volume replacement. Avoid in hyperkalemia (RL contains K)

WHAT IT ENCODES

LR carries ~4 mEq/L K⁺ and Plasma-Lyte ~5 mEq/L. In normal kidneys this is trivial and LR may even lower serum K⁺ (correcting acidosis shifts K⁺ intracellularly), but in hyperkalemic AKI/ESRD the prudent board answer is to avoid potassium-containing fluids and use NS.

BOARD TRAP

LR is dangerous in all renal failure because of its potassium. WRONG — the K⁺ load is small and often net-lowering via acidosis correction; the clean board contraindication is significant HYPERKALEMIA, where NS is preferred to avoid adding any K⁺.

5. D5W + NaHCO₃ (bicarbonate)

WHAT YOU SEE

D5W + NaHCO₃ (bicarbonate solution) — used in rhabdomyolysis to alkalinize urine and prevent pigment / Tamm-Horsfall cast formation

WHAT IT ENCODES

An isotonic bicarbonate drip is made by adding ~150 mEq NaHCO₃ to 1 L D5W. In rhabdomyolysis it alkalinizes the urine (goal urine pH >6.5) to reduce myoglobin–Tamm-Horsfall cast precipitation and limit pigment cast nephropathy; aggressive volume (target UOP 200–300 mL/h) is the more important intervention. Also used to alkalinize urine in salicylate or TCA toxicity and to treat severe metabolic acidosis.

BOARD TRAP

Bicarbonate fluid is a routine resuscitation fluid. WRONG — it is reserved for specific indications (rhabdo, severe acidemia, certain intoxications); overuse causes metabolic alkalosis, hypokalemia, hypocalcemia (alkalosis lowers ionized Ca, worsening rhabdo tetany), and volume overload.

6. Bicarb for metabolic acidosis

WHAT YOU SEE

Also used when metabolic acidosis needs correction

WHAT IT ENCODES

IV bicarbonate is reserved for severe acidemia (pH <7.1, especially with hemodynamic instability) and for non-anion-gap losses (diarrhea, RTA). In high-anion-gap lactic acidosis or DKA, bicarbonate is generally NOT given unless pH <7.0–7.1, since treating the cause regenerates bicarbonate and exogenous bicarb risks rebound alkalosis and paradoxical intracellular/CNS acidosis.

BOARD TRAP

Always give bicarbonate to correct DKA's low pH. WRONG — bicarbonate is avoided in DKA unless pH <7.0–7.1; insulin + fluids resolve the ketoacidosis and regenerate bicarbonate, and exogenous bicarb risks hypokalemia and cerebral edema (especially in children).

7. Packed RBCs

WHAT YOU SEE

Packed RBCs — severe acute blood loss

WHAT IT ENCODES

pRBCs restore oxygen-carrying capacity, not just volume. In massive hemorrhage use balanced resuscitation (~1:1:1 RBC:plasma:platelets) and permissive hypotension until bleeding is controlled; transfusion threshold is generally Hb <7 g/dL (<8 in ACS/cardiac disease). One unit raises Hb ~1 g/dL.

BOARD TRAP

Resuscitate active hemorrhagic shock with crystalloid alone. WRONG — large-volume crystalloid in hemorrhage causes dilutional coagulopathy, hypothermia, and acidosis (the 'lethal triad'); when blood is the loss, give blood products, not just saline.

8. Colloids (albumin)

WHAT YOU SEE

Colloids (albumin) — HRS: albumin infusion as bridge therapy. Not used routinely in AKI

WHAT IT ENCODES

Albumin (5% for volume, 25% for oncotic pull) stays intravascular longer than crystalloid per mL. Evidence-based niches: spontaneous bacterial peritonitis (1.5 g/kg day 1, 1 g/kg day 3 to prevent HRS), large-volume paracentesis (~6–8 g per L removed), and hepatorenal syndrome. The SAFE trial showed no general resuscitation benefit over saline (and possible harm in TBI).

BOARD TRAP

Albumin is a superior general resuscitation fluid. WRONG — SAFE showed equivalence to saline overall and harm in traumatic brain injury; albumin's role is narrow (SBP, LVP, HRS), not routine volume resuscitation.

9. Albumin in hepatorenal syndrome

WHAT YOU SEE

HRS: albumin as bridge therapy

WHAT IT ENCODES

HRS-AKI treatment = albumin (1 g/kg/day, max ~100 g) PLUS a splanchnic vasoconstrictor — terlipressin (first-line where available) or midodrine + octreotide, or norepinephrine in the ICU. Albumin expands central volume and binds vasoactive mediators; this combination is a bridge, not a cure.

BOARD TRAP

Vasoconstrictors alone treat HRS. WRONG — albumin must be combined with the vasoconstrictor (terlipressin or midodrine/octreotide); and pharmacotherapy is only a bridge — definitive treatment of HRS is liver transplantation.

10. HES increases AKI risk

WHAT YOU SEE

hydroxyethyl starch off the market — increased AKI risk

WHAT IT ENCODES

Synthetic hydroxyethyl starch colloids (Voluven, Hextend) increased AKI, need for renal replacement therapy, and mortality in critically ill/septic patients (VISEP, 6S, CHEST trials) and were withdrawn/restricted. Starch is deposited in renal tubules causing osmotic nephrosis.

BOARD TRAP

A starch colloid is a good volume expander for the septic patient. WRONG — HES is a classic exam 'harmful choice': it raises AKI, RRT, and death risk in critical illness and is off the market for these indications. Use balanced crystalloid.

11. The overarching rule

WHAT YOU SEE

The rule across all of them: match the fluid to what was lost and to the acid-base status of the patient

WHAT IT ENCODES

Fluid choice = composition of the loss + current acid-base/electrolyte state + goal (resuscitation vs maintenance). Maintenance ≈ '4-2-1' rule for rate, typically D5 1/2NS + 20 mEq KCl to cover free-water and glucose needs; resuscitation = isotonic balanced crystalloid boluses (e.g., 500 mL–1 L) titrated to perfusion. D5W (free water) is for hyponatremia/free-water deficit, NOT resuscitation (only ~1/12 stays intravascular).

BOARD TRAP

There is one best fluid for every patient. WRONG — boards test scenario-matching: hemorrhage → blood products; hyperchloremic acidosis → balanced crystalloid; rhabdo → isotonic fluids ± bicarbonate; hyponatremia/free-water deficit → D5W; hyperkalemia → potassium-free NS.